

PREFACE FOR FACULTY

Inversion of the Z-Transform is central to solving difference equations, deriving impulse responses, and synthesizing digital filter structures. This lecture provides a comprehensive treatment of all three inversion techniques: **Partial Fraction Expansion (PFE)**, **Power Series Expansion (Long Division)**, and the **Contour Inversion Integral / Cauchy Residue Theorem**.

Pedagogical Strategy:

1. Master Partial Fraction Expansion for distinct real poles, repeated poles, and complex conjugate pole pairs.
 2. Teach the Long Division method for obtaining the first N samples of non-rational or high-order transfer functions.
 3. Review the Cauchy Residue Theorem for complex contour integration along a closed counterclockwise path in the ROC: $x[n] = \frac{1}{2\pi j} \oint_C X(z)z^{n-1}dz = \sum \text{Res}[X(z)z^{n-1}]$.
 4. Link the pole locations of the rational transfer function $H(z) = \frac{B(z)}{A(z)}$ directly to transient decay rates and sinusoidal oscillation modes.
 5. System stability and causality criteria from pole-zero diagrams.
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1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Apply** Partial Fraction Expansion to compute $x[n]$ for distinct, repeated, and complex poles under different ROC constraints.
 2. **Execute** polynomial long division to extract sample values for causal ($|z| > r_0$) and anti-causal ($|z| < r_0$) sequences.
 3. **Evaluate** inverse transforms using the Cauchy Residue Theorem.
 4. **Solve** Linear Constant-Coefficient Difference Equations (LCCDE) with non-zero initial conditions using the One-Sided (Unilateral) Z-Transform.
 5. **Determine** stability, causality, and frequency selectivity from pole-zero plots.
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2. MATHEMATICAL FOUNDATIONS

2.1 Inversion Methods Overview

1. **Partial Fraction Expansion (PFE):** Most practical for rational $X(z) = \frac{B(z)}{A(z)}$.
2. **Power Series (Long Division):** Expands $X(z) = \sum x[n]z^{-n}$ directly.
3. **Contour Inversion Integral (Cauchy Residue Theorem):**

$$x[n] = \frac{1}{2\pi j} \oint_C X(z)z^{n-1}dz = \sum_{\text{poles inside } C} \text{Res} [X(z)z^{n-1}]$$

Where the residue at a simple pole p_k is: $\text{Res}\{X(z)z^{n-1}\}_{z=p_k} = \lim_{z \rightarrow p_k} (z - p_k) X(z)z^{n-1}$

2.2 Partial Fraction Expansion Formulation

Given $X(z) = \frac{b_0 + b_1 z^{-1} + \dots + b_M z^{-M}}{1 + a_1 z^{-1} + \dots + a_N z^{-N}}$. It is mathematically cleanest to expand $\frac{X(z)}{z}$ in terms of z :

$$\frac{X(z)}{z} = \frac{A_1}{z - p_1} + \frac{A_2}{z - p_2} + \dots + \frac{A_N}{z - p_N}$$

Where residue $A_k = \lim_{z \rightarrow p_k} (z - p_k) \frac{X(z)}{z}$. Then: $X(z) = \sum_{k=1}^N A_k \frac{z}{z - p_k} = \sum_{k=1}^N A_k \frac{1}{1 - p_k z^{-1}}$. Inversion depends on the ROC for each pole p_k :

- If ROC is outside pole ($|z| > |p_k|$): $A_k(p_k)^n u[n]$ (Causal).
- If ROC is inside pole ($|z| < |p_k|$): $-A_k(p_k)^n u[-n - 1]$ (Anti-causal).

3. WORKED NUMERICAL EXAMPLES

Example 6.1: Inversion under Different ROC Constraints

Problem: Find $x[n]$ for $X(z) = \frac{1}{(1 - 0.5z^{-1})(1 - 2z^{-1})}$ for: (a) ROC: $|z| > 2$ (Causal). (b) ROC: $|z| < 0.5$ (Anti-causal). (c) ROC: $0.5 < |z| < 2$ (Two-sided).

Solution: Partial fraction expansion in z^{-1} :

$$X(z) = \frac{A}{1 - 0.5z^{-1}} + \frac{B}{1 - 2z^{-1}}$$

$$A = \lim_{z \rightarrow 2} (1 - 0.5z^{-1}) X(z) \big|_{z^{-1} = 2} = \frac{1}{1 - 2(2)} = -\frac{1}{3}$$

$$B = \lim_{z \rightarrow 0.5} (1 - 2z^{-1}) X(z) \big|_{z^{-1} = 0.5} = \frac{1}{1 - 0.5(0.5)} = \frac{4}{3}$$

$$X(z) = -\frac{1}{3} \frac{1}{1 - 0.5z^{-1}} + \frac{4}{3} \frac{1}{1 - 2z^{-1}}$$

- **(a) ROC: $|z| > 2$:** Both poles are inside ROC $\implies$ Both terms causal:

$$x[n] = -\frac{1}{3}(0.5)^n u[n] + \frac{4}{3}(2)^n u[n]$$

- **(b) ROC: $|z| < 0.5$:** Both poles are outside ROC $\implies$ Both terms anti-causal:

$$x[n] = \frac{1}{3}(0.5)^n u[-n - 1] - \frac{4}{3}(2)^n u[-n - 1]$$

- **(c) ROC: $0.5 < |z| < 2$:** $|z| > 0.5$ (causal for $p = 0.5$) and $|z| < 2$ (anti-causal for $p = 2$):

$$x[n] = -\frac{1}{3}(0.5)^n u[n] - \frac{4}{3}(2)^n u[-n - 1]$$

4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

Solve the difference equation with initial conditions using the unilateral Z-transform:

$$y[n] - 0.7y[n - 1] + 0.1y[n - 2] = x[n], \quad n \geq 0$$

Given $x[n] = u[n]$ and initial conditions $y[-1] = 1, y[-2] = 2$. (15 Marks)

Model Answer & Step-by-Step Marking Rubric:

- **Unilateral Z-transform property:** $\mathcal{Z}y[n-1] = z^{-1}Y(z) + y[-1]$ $\mathcal{Z}y[n-2] = z^{-2}Y(z) + z^{-1}y[-1] + y[-2]$ (3 Marks)
- **Transform difference equation:** $[Y(z)] - 0.7[z^{-1}Y(z) + 1] + 0.1[z^{-2}Y(z) + z^{-1}(1) + 2] = \frac{1}{1-z^{-1}}$
 $Y(z)[1 - 0.7z^{-1} + 0.1z^{-2}] - 0.7 + 0.1z^{-1} + 0.2 = \frac{1}{1-z^{-1}}$
 $Y(z)[1 - 0.7z^{-1} + 0.1z^{-2}] = 0.5 - 0.1z^{-1} + \frac{1}{1-z^{-1}} = \frac{(0.5-0.1z^{-1})(1-z^{-1})+1}{1-z^{-1}} = \frac{1.5-0.6z^{-1}+0.1z^{-2}}{1-z^{-1}}$ (4 Marks)
- **Factor denominator:** $1 - 0.7z^{-1} + 0.1z^{-2} = (1 - 0.5z^{-1})(1 - 0.2z^{-1})$.

$$Y(z) = \frac{1.5 - 0.6z^{-1} + 0.1z^{-2}}{(1 - z^{-1})(1 - 0.5z^{-1})(1 - 0.2z^{-1})}$$

(3 Marks)

- **Partial Fraction Expansion:**

$$Y(z) = \frac{A}{1 - z^{-1}} + \frac{B}{1 - 0.5z^{-1}} + \frac{C}{1 - 0.2z^{-1}}$$

$$\begin{aligned} \circ A &= \left. \frac{1.5-0.6(1)+0.1(1)}{(1-0.5)(1-0.2)} \right|_{z^{-1}=1} = \frac{1.0}{0.5 \times 0.8} = \frac{1.0}{0.4} = 2.5. \\ \circ B &= \left. \frac{1.5-0.6(2)+0.1(4)}{(1-2)(1-0.4)} \right|_{z^{-1}=2} = \frac{1.5-1.2+0.4}{(-1)(0.6)} = \frac{0.7}{-0.6} = -1.1667 = -\frac{7}{6}. \\ \circ C &= \left. \frac{1.5-0.6(5)+0.1(25)}{(1-5)(1-2.5)} \right|_{z^{-1}=5} = \frac{1.5-3.0+2.5}{(-4)(-1.5)} = \frac{1.0}{6.0} = \frac{1}{6}. \text{ (3 Marks)} \end{aligned}$$

- **Inverse Transform:**

$$y[n] = \left[2.5 - \frac{7}{6}(0.5)^n + \frac{1}{6}(0.2)^n \right] u[n]$$

(2 Marks)

5. PYTHON VERIFICATION SCRIPT

```
import numpy as np

# Simulate LCCDE iteratively
N = 10
y = np.zeros(N + 2)
y[0] = 2 # y[-2]
y[1] = 1 # y[-1]

for n in range(2, N + 2):
    y[n] = 0.7 * y[n-1] - 0.1 * y[n-2] + 1.0

n_idx = np.arange(0, N)
y_sim = y[2:]
y_analytic = 2.5 - (7/6)*(0.5**n_idx) + (1/6)*(0.2**n_idx)

print("Simulated y[n]:", np.round(y_sim[:5], 4))
print("Analytic y[n]: ", np.round(y_analytic[:5], 4))
```